

Software Test Design (STD) – Ambulo App

This document contains the Software Test Design for the Ambulo Hiking App. It includes detailed test cases derived from the app's test code files, organized by feature and functionality.

Test Case ID	Description	Preconditions	Test Steps	Expected Result
TC-AUTH-001	Register user with valid email and password	App running, internet connection	1. Open file: dataManagerManualTests.dart 2. Call function: _runAuthTest() 3. Check for log: 'Registration successful: ...'	User is registered and authenticated
TC-AUTH-002	Login with registered credentials	User already registered	1. Open file: dataManagerManualTests.dart 2. Call function: _runAuthTest() 3. Observe step: signIn() is called 4. Confirm log: 'Sign in successful: ...'	User is logged in successfully
TC-AUTH-003	Logout user	User logged in	1. Open file: dataManagerManualTests.dart 2. Call function: _runAuthTest() 3. Observe signOut() is triggered 4. Confirm user session is cleared	User is signed out
TC-AUTH-004	Delete user account	User logged in	1. Open file: dataManagerManualTests.dart 2. Call function: _runAuthTest() 3. Observe deleteUser() called 4. Confirm log: 'User successfully deleted'	User is deleted and cannot log in again
TC-PREF-001	Create user profile after registration	User registered	1. Open file: dataManagerManualTests.dart	User profile is created in Firestore

			2. Call function: <code>_runUserTest()</code> 3. Confirm <code>createUserProfile()</code> is called	
TC-PREF-002	Retrieve user preferences	User profile exists	1. Open file: <code>dataManagerManualTests.dart</code> 2. Call function: <code>_runUserTest()</code> 3. Confirm <code>getUserPreferences()</code> returns preferences	User preferences retrieved successfully
TC-PREF-003	Update user theme preference	User logged in	1. Open file: <code>dataManagerManualTests.dart</code> 2. Call function: <code>_runUserTest()</code> 3. Call: <code>updatePreferences(userId, 'theme', 'dark')</code>	User theme set to dark mode
TC-PREF-004	Check default language in preferences	User preferences created	1. Open file: <code>dataManagerManualTests.dart</code> 2. Observe default value of 'language' in preferences	Default language is set (e.g., 'en')
TC-PREF-005	Add trail to hiking history	Trail created, user exists	1. Open file: <code>dataManagerManualTests.dart</code> 2. Function: <code>_runUserTest()</code> 3. Calls: <code>createTrail()</code> and <code>addTrailToHistory()</code>	Trail appears in user's hiking history
TC-TRAIL-001	Create trail using form input	App running, user authenticated	1. Open file: <code>trailTest.dart</code> 2. Fill form fields: name, gpx, description, distance, etc. 3. Press 'Create Trail' button	Trail is created and ID is returned
TC-TRAIL-002	Edit trail details	Trail exists	1. Open file: <code>trailTest.dart</code> 2. Click 'Edit Trail Details' 3. Enter key and value 4. Confirm update in logs	Trail details updated successfully
TC-TRAIL-003	Update trail description	Trail selected	1. Open file: <code>trailTest.dart</code> 2. Click 'Write Description' 3. Enter new description 4. Confirm update	Trail description updated
TC-TRAIL-004	Update trail rating	Trail exists	1. Open file: <code>trailTest.dart</code> 2. Click 'Update Rating' 3. Enter numeric rating and confirm	Trail rating updated in Firestore

TC-TRAIL-005	Delete a trail	Trail exists	1. Open file: trailTest.dart 2. Click 'Delete Trail' 3. Confirm deletion	Trail deleted successfully
TC-MEDIA-001	Upload profile picture manually	User registered	1. Open file: dataManagerManualTests.dart 2. Call _uploadProfilePictureTest() 3. Observe 'Profile picture uploaded successfully'	Image is uploaded and URL stored
TC-MEDIA-002	Upload trail image and reload	Trail exists	1. Open file: dataManagerImageTests.dart 2. Click green IconButton in AppBar 3. Confirm images are reloaded	Trail image uploaded and previewed
TC-MEDIA-003	Delete trail image	Image exists	1. Open file: dataManagerManualTests.dart 2. Call _uploadTrailImageTest() 3. Observe deletion log after upload	Image deleted and not shown
TC-WEATHER-001	Fetch forecast for Tel Aviv	Internet available	1. Open file: WeatherTest.dart 2. Click button labeled 'Tel Aviv' 3. View forecast widgets and logs	Weather data shown for Tel Aviv
TC-WEATHER-002	Get current location and fetch forecast	Location permission granted	1. Open file: WeatherTest.dart 2. Click 'Use Current Location' 3. Wait for forecast to load	Weather displayed for current coordinates
TC-UI-001	Navigate to UserTestsPage	User registered	1. Open file: shai_page.dart 2. Click 'Go to User Tests' button 3. Confirm navigation to UserTestsPage	UserTestsPage is displayed
TC-UI-002	Navigate to MapPage	None	1. Open file: shai_page.dart 2. Click 'Go To Map Page' button	MapPage is displayed
TC-UI-003	Navigate to Saved Routes	User logged in	1. Open file: shai_page.dart 2. Click 'Go to Saved Routes Page'	SavedRoutesPage is shown with user's trails
TC-TRAILPAG	Load TrailPage	trailId and testUser	1. Open file: TrailPageTest.dart	TrailPage displays trail

E-001	with given trailId and user	provided	2. Pass trailId and testUser to TrailPage 3. Confirm TrailPage UI is rendered	info for user
TC-USER-001	Create and load user profile	App running	1. Open file: userTests.dart 2. Click 'Create and Load' 3. Check log: 'User created and loaded'	User profile is created and state loaded
TC-USER-002	Toggle theme preference	User logged in	1. Open file: userTests.dart 2. Click 'Toggle Theme' button	User theme setting toggled
TC-USER-003	Add completed hike	User logged in	1. Open file: userTests.dart 2. Click 'Complete Trail' and enter trail ID	Trail marked as completed for user
TC-UI-004	DanielPage loads LoginPage	None	1. Open file: daniel_page.dart	LoginPage is displayed
TC-UI-005	Show all trails using AllTrailsWidget	Trails available in DB	1. Open file: show_all_trail_test.dart	AllTrailsWidget renders list of trails